Faster, Cheaper, or Safer?

The engineering trade-off triangle

SCAN / LEARN

THE MAIN IDEA

Every design trades cost, speed and reliability. Optimization starts with a measurable goal, then a test of both benefits and side effects.

FOLLOW THE SYSTEM

01 MEASURE
Record speed, errors and resource cost.

02 CHOOSE
Select one bottleneck worth improving.

03 CHANGE
Try a cache, smaller worker or better route.

04 COMPARE
Measure the new result and its risks.

MEASURE → CHANGE → MEASURE AGAIN → DECIDE

COMMON MISCONCEPTION

The cheapest solution is automatically the most efficient.

MORE ACCURATE

A cheap failure can cost more than a reliable service.

THREE QUESTIONS TO KEEP

PROBLEM: How do we spend fewer resources safely?

FAILURE: Optimizing one number makes another worse.

IN REAL LIFE: Cloud bills, AI inference and streaming.

SOLO DISCOVERY

Compare two designs: one faster but 2x costlier. Choose a goal.